# Auditing Membership Leakage in Small SMILES Language Models: LiRA, Randomized-SMILES Training and DP-SGD (Preliminary Study)

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Abstract—Autoregressive language models trained on molecular string representations (SMILES) are increasingly explored for computer-aided molecular generation. However, training sets in biopharmaceutical contexts often contain proprietary chemical structures, confidential screening hits, and patented scaffolds. In this study, we empirically evaluate Membership Inference Attacks (MIA) against small causal SMILES language models (2 layers, $d = 128$, $\sim 0.4$M parameters) on a subset of the MOSES benchmark. We evaluate standard sequence loss thresholding (LOSS) alongside calibrated Likelihood-Ratio Attacks (LiRA) across training epochs (5, 20, 60), dataset sizes ($N \in \{1\text{k}, 2.5\text{k}, 5\text{k}, 10\text{k}\}$), a transfer learning scenario (pretrain then fine-tune), and two defensive strategies: randomized-SMILES data augmentation and Differentially Private SGD (DP-SGD).
In our baseline setting evaluated with canonical SMILES queries and sequence negative log-likelihood sum ($nll_sum$), models exhibit substantial membership leakage by epoch 60: online LiRA reaches a ROC-AUC of $0.898 \pm 0.003$ and a True Positive Rate (TPR) of $27.82 \pm 1.47\%$ at a 1% False Positive Rate (FPR) ($8.15 \pm 1.58\%$ at 0.1% FPR), compared to $0.689 \pm 0.005$ AUC and $2.96 \pm 0.24\%$ TPR@1% FPR for standard LOSS. In smaller training cohorts ($N = 1,000$), attack sensitivity increases (LOSS AUC reaches $0.998 \pm 0.001$, TPR@1% $= 93.99 \pm 4.35\%$; offline LiRA AUC reaches $0.957 \pm 0.006$, TPR@1% $= 20.75 \pm 6.51\%$; online LiRA AUC reaches $0.975 \pm 0.004$, TPR@1% $= 45.49 \pm 33.14\%$, with high LiRA estimation variance). Private fine-tuning after public pretraining also exhibits leakage (offline LiRA AUC $0.801 \pm 0.003$, TPR@1% $= 4.36 \pm 0.51\%$).
In our setting, mean-aggregated scoring over up to 10 string forms (canonical included) removes most of the signal: querying the epoch-60 canonical baseline with up to 10 distinct SMILES strings per molecule (the canonical form plus randomized forms, with mean NLL) reduces online LiRA AUC to $0.528 \pm 0.006$ (offline LiRA AUC $0.560 \pm 0.007$, LOSS AUC $0.511 \pm 0.004$). We did not score individual non-canonical strings, so we cannot say whether single randomized forms leak. Training with randomized-SMILES augmentation similarly suppresses canonical attack AUC to $0.528 \pm 0.004$. In our preliminary DP-SGD arm ($\epsilon \in \{8.0, 3.0\}$, $\delta = 10^{-5}$), attack AUC drops to $0.500 - 0.528$, with possible loss of generative utility: an attack AUC near 0.5 may reflect an under-trained model rather than an effective defense, as generative validity was not evaluated. Finally, an empirical audit of 15,000 generated samples across baseline models observed 12 exact member replications versus 7 holdout replications; a one-sided binomial test ($p = 0.1796 > 0.05$) and Fisher exact test ($p = 0.359 > 0.05$) indicate that this difference is not statistically significant at the current sample size. We conclude with a discussion of open limitations and recommendations for privacy evaluation in chemical language models.

Index Terms—Membership Inference Attacks, Molecular Generative Models, SMILES, Differential Privacy, Likelihood-Ratio Attack, Privacy Auditing.

## 1 Introduction

Autoregressive generative sequence models parameterize discrete token distributions to explore chemical space. By modeling the Simplified Molecular Input Line Entry System (SMILES) [1], causal transformers can generate candidate structures and propose chemical variations for lead optimization.

In commercial drug discovery, generative models are often fine-tuned or trained directly on proprietary compound collections, including patented chemical series, internal bioassay screening hits, and confidential leads. If an adversary can infer whether a specific target compound was included in the training cohort—a vulnerability formalized as a Membership Inference Attack (MIA) [2]—they may extract competitive intelligence regarding active medicinal chemistry programs or unreleased intellectual property.

While membership inference has been widely investigated in computer vision and natural language processing, auditing molecular generative sequence models presents distinctive factors:

1) Molecules admit multiple non-unique equivalent SMILES representations (non-canonical string enumeration).
2) Chemical libraries often contain recurring core frameworks (Bemis-Murcko scaffolds [3]) that induce structural correlation across compounds.
3) Language models may memorize specific surface syntax without learning generalized semantic invariants.

In this work, we present an empirical audit of membership leakage in small SMILES autoregressive models. Using a shadow-model ensemble framework based on Carlini et al.'s Likelihood-Ratio Attack (LiRA) [4], we evaluate exposure

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across training duration, dataset scaling, transfer learning, and defensive interventions.

## 2 Threat Model and Formulation

### 2.1 Sequence Modeling and Scoring Metric

Let a molecular SMILES string be represented as a token sequence $x = (x_1, \ldots, x_T) \in \mathcal{V}^T$, where $\mathcal{V}$ is a discrete vocabulary of chemical tokens. An autoregressive transformer parameterizes the likelihood:

$$P_\theta(x) = \prod_{t=1}^{T} P_\theta(x_t \mid x_{<t}) \tag{1}$$

The sequence negative log-likelihood (NLL) sum is defined as:

$$\mathrm{NLL}_{\mathrm{sum}}(\theta; x) = -\sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t}) \tag{2}$$

Throughout this paper, we consistently report and evaluate metrics using sequence NLL sum (*nll_sum*). In our experiments, *nll_sum* and *nll_mean* yield near-identical LiRA results across epochs 5, 20, and 60 (LiRA z-scores are mathematically invariant to per-molecule length scaling), while uncalibrated LOSS differs only marginally ($\Delta\mathrm{AUC} \leq 0.004$); we report *nll_sum* consistently throughout. Where indicated, we also report randomized-SMILES variant scoring (*nll_variants_mean*), which averages NLL across up to 10 distinct SMILES strings per molecule (the canonical form plus randomized forms), under mean aggregation (individual non-canonical forms were not scored separately).

### 2.2 Attack Formulations: LOSS vs. LiRA

Given a query molecule $x$, the adversary computes a test statistic $S(x)$ to predict whether $x \in \mathcal{D}_{\mathrm{train}}$:

1) Loss Thresholding (LOSS):

$$S_{\mathrm{LOSS}}(x) = -\mathrm{NLL}_{\mathrm{sum}}(\mathcal{M}_{\mathrm{target}}; x) \tag{3}$$

LOSS assumes that training examples achieve lower sequence cross-entropy loss than non-training examples. However, intrinsic molecular complexity varies substantially: complex macrocycles or long branched chains exhibit higher baseline loss than simple aromatics regardless of membership.

2) Calibrated Likelihood-Ratio Attack (LiRA): To adjust for molecular complexity, LiRA [4] trains an ensemble of $K$ shadow models $\{\mathcal{M}_k\}_{k=1}^{K}$ on random subsets of the data pool. For each query $x$, the models are partitioned into in-models $\mathcal{S}_{\mathrm{in}}(x) = \{k : x \in \mathcal{D}_k\}$ and out-models $\mathcal{S}_{\mathrm{out}}(x) = \{k : x \notin \mathcal{D}_k\}$.

   - Offline LiRA (Out-only): Estimates the out-distribution parameters $\mu_{\mathrm{out}}(x)$ and $\sigma_{\mathrm{out}}(x)$ over out-models:

$$\Lambda_{\mathrm{offline}}(x) = -\frac{\mathrm{NLL}_{\mathrm{sum}}(\mathcal{M}_{\mathrm{target}}; x) - \mu_{\mathrm{out}}(x)}{\sigma_{\mathrm{out}}(x) + 10^{-6}} \tag{4}$$

   - Online LiRA (In vs. Out): Estimates both in-distribution $(\mu_{\mathrm{in}}, \sigma_{\mathrm{in}})$ and out-distribution $(\mu_{\mathrm{out}}, \sigma_{\mathrm{out}})$ parameters, computing the log-likelihood ratio under parametric Gaussian assumptions:

$$\begin{aligned}\Lambda_{\mathrm{online}}(x) = &-\frac{1}{2}\left(\frac{\mathrm{obs} - \mu_{\mathrm{in}}}{\sigma_{\mathrm{in}}}\right)^2 \\ &+ \frac{1}{2}\left(\frac{\mathrm{obs} - \mu_{\mathrm{out}}}{\sigma_{\mathrm{out}}}\right)^2 \\ &+ \log(\sigma_{\mathrm{out}}/\sigma_{\mathrm{in}})\end{aligned} \tag{5}$$

### 2.3 Threat Model Assumptions

We evaluate our models under an explicit, optimistic threat model:

1) Distribution Match: Shadow models are sampled from the exact same marginal distribution as the target model (MOSES benchmark pool).
2) Likelihood Access: The adversary has query access enabling exact calculation of token-level sequence log-likelihoods (*nll_sum*).
3) Canonical-String Queries: The adversary queries the target and shadow models using standard canonical SMILES representations.

These represent strong assumptions favoring the attacker. In restricted black-box deployments (e.g., API rate limits, generation-only access, or query filtering), adversary effectiveness may be substantially lower.

## 3 Experimental Setup and Architecture

### 3.1 Model Architecture and Training Setup

All evaluations utilize a small causal GPT architecture: 2 transformer layers, hidden embedding dimension $d_{\mathrm{model}} = 128$, 4 attention heads, dropout 0.1, and a context window of 128 tokens. Instantiating the model over the dataset tokenizer (len(tokenizer.itos)) yields an exact vocabulary size of 25 tokens and 419,584 parameters. Models are trained using AdamW (lr $= 10^{-3}$, weight_decay $= 0.01$), batch size 32, with gradient clipping norm 1.0 and a fixed sequence length of 128 tokens. Differentially private models (DPCausalGPT) evaluate the identical architecture (419,584 parameters) and utilize the same optimizer configuration (AdamW, lr $= 10^{-3}$, weight_decay $= 0.01$, batch size 32, dropout 0.1, max_grad_norm $= 1.0$) wrapped via Opacus.

### 3.2 Dataset and Shadow Model Partitioning

Data are sourced from the MOSES benchmark [5]. The evaluation pool size is POOL_N $= 20,000$ molecules. For each shadow model $k$, exactly $N = 10,000$ molecules are sampled uniformly at random without replacement using seed $k$ (_membership(pool, k) in colab_pipeline.py). Evaluation is conducted across all 20,000 candidate compounds (10,000 members, 10,000 non-members). In the baseline and randomized-augmentation (E5a) experiments, 32 shadow models are trained ($k \in \{0, \ldots, 31\}$). In data scaling (E3), DP-SGD, and pretrain-finetune experiments, 8 shadow models are trained per condition.

### 3.3 Opacus DP-SGD Implementation

Differentially private training is implemented using Opacus (version 1.6.0) [6]. The noise scale $\sigma$ is computed dynamically using opacus.accountants.utils.get_noise_multiplier(target_epsilon=eps,target_delta=1e-5,sample_rate=32/N,epochs=60). Gradient clipping is set to max_grad_norm = 1.0, targeting $\epsilon \in \{8.0, 3.0\}$ with $\delta = 10^{-5}$. The causal attention layer avoids fused operations incompatible with Opacus per-sample gradient hooks. If training resumes from a checkpoint, privacy accounting restarts from zero unless serialized accountant state is restored. In results_summary/manifest.json, a stale launch record with an ImportError for DPCausalGPT from an earlier initialization attempt is noted; subsequent runs completed.

## 4 Empirical Results and Findings

### 4.1 Baseline Leakage vs. Training Duration

Table 1 and Fig. 1 present membership inference performance across 5, 20, and 60 training epochs on canonical SMILES representations evaluated using *nll_sum*.

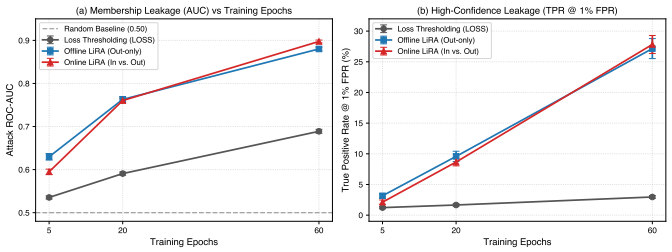



Fig. 1. Baseline Attack Vulnerability vs. Training Epochs. (a) ROC-AUC progression. (b) True Positive Rate at 1% False Positive Rate.

At epoch 5, membership leakage is modest (strongest attack: offline LiRA AUC 0.630 ± 0.008). By epoch 60, extensive exposure occurs: online LiRA reaches ROC-AUC 0.898±0.003, with TPR@1% FPR reaching 27.82±1.47%. At a stricter threshold of 0.1% FPR, online LiRA still identifies 8.15±1.58% of member molecules. Standard loss thresholding substantially underestimates vulnerability (AUC 0.689 vs. 0.898 for online LiRA at epoch 60).

Statistical Uncertainty: The reported ± values reflect the standard deviation across 32 leave-one-out shadow target models. To assess uncertainty over the 20,000 evaluation molecules, we computed 95% bootstrap confidence intervals for epoch-60 online LiRA across three target models (not independent: they share shadow models and overlapping training pools):

- Target 0: AUC 0.9004, 95% CI $[0.8962, 0.9045]$; TPR@1% 0.2684, 95% CI $[0.2492, 0.3000]$.
- Target 1: AUC 0.8916, 95% CI $[0.8870, 0.8961]$; TPR@1% 0.2510, 95% CI $[0.2178, 0.2756]$.
- Target 2: AUC 0.8972, 95% CI $[0.8930, 0.9012]$; TPR@1% 0.2838, 95% CI $[0.2613, 0.3136]$.

### 4.2 String-Form Sensitivity: Randomized-SMILES Scoring

In our setting, mean-aggregated scoring over up to 10 string forms (canonical included) removes most of the signal. In colab_pipeline.py, each molecule was scored at epoch 60 using up to 10 distinct SMILES strings per molecule (the canonical form plus randomized forms generated via RDKit Chem.MolToSmiles(mol, doRandom=True, canonical=False)), with the average NLL recorded as *nll_variants_mean*.

As reported in Table 1 (bottom section), scoring canonical models with randomized variants drops LOSS AUC from 0.689 to 0.511 ± 0.004, offline LiRA AUC from 0.880 to 0.560 ± 0.007, and online LiRA AUC from 0.898 to 0.528 ± 0.006. We did not score individual non-canonical strings, so we cannot say whether single randomized forms leak.

### 4.3 Private Dataset Scaling

We examine vulnerability across dataset sizes $N \in \{1\text{k}, 2.5\text{k}, 5\text{k}, 10\text{k}\}$ trained for 60 epochs, with 8 shadow models per size ($2N$ candidate universe). Results are presented in Table 2 and Fig. 2, separating LOSS and LiRA attacks.

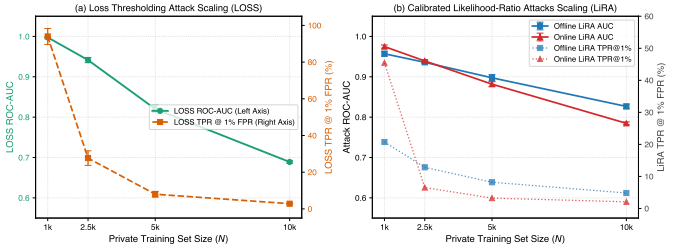



Fig. 2. Dataset Size Scaling (E3). (a) Loss thresholding attack (LOSS). (b) Calibrated Likelihood-Ratio Attacks (LiRA).

Smaller training cohorts experience pronounced overfitting at 60 epochs. At $N = 1,000$, standard LOSS reaches AUC 0.998 ± 0.001 with TPR@1% = 93.99 ± 4.35%, while offline LiRA reaches AUC 0.957 ± 0.006 (TPR@1% = 20.75 ± 6.51%) and online LiRA reaches AUC 0.975 ± 0.004 (TPR@1% = 45.49 ± 33.14%). With only 8 shadow models, LiRA density estimation at small $N$ suffers from high variance (e.g., online LiRA TPR@1% standard deviation is ±33.14%). We report these empirical numbers directly and do not claim a formal parametric scaling law.

### 4.4 Transfer Learning: Pretraining to Fine-Tuning

We evaluated a transfer learning setting where a model was pretrained on 10,000 public molecules for 60 epochs, then fine-tuned on 10,000 private molecules for 20 epochs across 8 shadow targets (Table 3).

Offline LiRA achieves an AUC of 0.801 ± 0.003 and TPR@1% of 4.36 ± 0.51%. While lower than baseline epoch 60 (0.880 ± 0.004 / 27.15 ± 1.63%), offline LiRA AUC is higher than the baseline trained from scratch for 20 epochs (0.763 ± 0.007 / 9.58 ± 0.84%, though TPR@1% is lower), indicating that fine-tuning on private data remains susceptible to membership inference. However, there is no matched from-scratch control trained on the identical 10,000 private compounds for 20 epochs without pretraining, so the isolated effect of pretraining cannot be disentangled.

TABLE 1
Baseline membership inference vulnerability across training epochs on canonical SMILES. Evaluated over 32 shadow models on 20,000 candidate targets (10,000 members, 10,000 non-members). Values report mean ± standard deviation across 32 leave-one-out target evaluations (not a confidence interval). All metrics use negative log-likelihood sum (nll_sum), with randomized-SMILES variants scoring (up to 10 distinct SMILES strings per molecule, the canonical form plus randomized forms, mean NLL) reported at epoch 60.

| Epochs | Scoring Representation | Attack Method | ROC-AUC | TPR @ 1% FPR (%) | TPR @ 0.1% FPR (%) |
|---|---|---|---|---|---|
| 5 | Canonical (nll_sum) | Loss Thresholding (LOSS) | $0.535 \pm 0.004$ | $1.23 \pm 0.18$ | $0.14 \pm 0.07$ |
| 5 | Canonical (nll_sum) | Offline LiRA (Out-only) | $0.630 \pm 0.008$ | $3.15 \pm 0.36$ | $0.42 \pm 0.15$ |
| 5 | Canonical (nll_sum) | Online LiRA (In vs. Out) | $0.595 \pm 0.006$ | $2.12 \pm 0.35$ | $0.19 \pm 0.09$ |
| 5 | Canonical (nll_sum) | Strongest: offline_LiRA | $0.630 \pm 0.008$ | $3.15 \pm 0.36$ | $0.42 \pm 0.15$ |
| 20 | Canonical (nll_sum) | Loss Thresholding (LOSS) | $0.591 \pm 0.004$ | $1.66 \pm 0.21$ | $0.21 \pm 0.08$ |
| 20 | Canonical (nll_sum) | Offline LiRA (Out-only) | $0.763 \pm 0.007$ | $9.58 \pm 0.84$ | $1.86 \pm 0.47$ |
| 20 | Canonical (nll_sum) | Online LiRA (In vs. Out) | $0.760 \pm 0.006$ | $8.66 \pm 0.62$ | $1.34 \pm 0.33$ |
| 20 | Canonical (nll_sum) | Strongest: offline_LiRA | $0.763 \pm 0.007$ | $9.58 \pm 0.84$ | $1.86 \pm 0.47$ |
| 60 | Canonical (nll_sum) | Loss Thresholding (LOSS) | $0.689 \pm 0.005$ | $2.96 \pm 0.24$ | $0.41 \pm 0.15$ |
| 60 | Canonical (nll_sum) | Offline LiRA (Out-only) | $0.880 \pm 0.004$ | $27.15 \pm 1.63$ | $8.22 \pm 1.32$ |
| 60 | Canonical (nll_sum) | Online LiRA (In vs. Out) | $0.898 \pm 0.003$ | $27.82 \pm 1.47$ | $8.15 \pm 1.58$ |
| 60 | Canonical (nll_sum) | Strongest: online_LiRA | $0.898 \pm 0.003$ | $27.82 \pm 1.47$ | $8.15 \pm 1.58$ |
| 60 (variants) | Randomized-SMILES (up to 10 strings, mean NLL) | Loss Thresholding (LOSS) | $0.511 \pm 0.004$ | $1.09 \pm 0.12$ | $0.12 \pm 0.05$ |
| 60 (variants) | Randomized-SMILES (up to 10 strings, mean NLL) | Offline LiRA (Out-only) | $0.560 \pm 0.007$ | $2.01 \pm 0.29$ | $0.30 \pm 0.10$ |
| 60 (variants) | Randomized-SMILES (up to 10 strings, mean NLL) | Online LiRA (In vs. Out) | $0.528 \pm 0.006$ | $1.45 \pm 0.47$ | $0.18 \pm 0.12$ |
| 60 (variants) | Randomized-SMILES (up to 10 strings, mean NLL) | Strongest: offline_LiRA | $0.560 \pm 0.007$ | $2.01 \pm 0.29$ | $0.30 \pm 0.10$ |

TABLE 2
Membership leakage scaling as private dataset size varies ($N \in \{1k, 2.5k, 5k, 10k\}$). Each size uses 8 shadow models ($2N$ candidate universe). LiRA estimates exhibit high variance at small $N$ due to limited shadow samples per molecule. Evaluated using nll_sum. We present empirical measurements without assuming a parametric power law.

| | Loss Thresholding (LOSS) | | Offline LiRA | | Online LiRA | |
|---|---|---|---|---|---|---|
| Size ($N$) | ROC-AUC | TPR@1% (%) | ROC-AUC | TPR@1% (%) | ROC-AUC | TPR@1% (%) |
| 1,000 | $0.998 \pm 0.001$ | $93.99 \pm 4.35$ | $0.957 \pm 0.006$ | $20.75 \pm 6.51$ | $0.975 \pm 0.004$ | $45.49 \pm 33.14$ |
| 2,500 | $0.941 \pm 0.006$ | $27.68 \pm 3.97$ | $0.936 \pm 0.003$ | $12.85 \pm 2.97$ | $0.939 \pm 0.003$ | $6.53 \pm 1.60$ |
| 5,000 | $0.820 \pm 0.010$ | $7.99 \pm 1.55$ | $0.898 \pm 0.008$ | $8.19 \pm 1.05$ | $0.882 \pm 0.006$ | $3.23 \pm 0.60$ |
| 10,000 | $0.689 \pm 0.003$ | $2.84 \pm 0.09$ | $0.827 \pm 0.006$ | $4.81 \pm 0.63$ | $0.785 \pm 0.003$ | $2.07 \pm 0.28$ |

TABLE 3
Membership leakage in transfer learning (public pretraining on 10,000 molecules, fine-tuning on 10,000 private molecules). Metrics report mean ± std across 8 target models (not independent: they share shadow models and overlapping training pools). Note: there is no from-scratch control trained at the exact same size/epochs in this experiment.

| Attack Method | ROC-AUC | TPR @ 1% FPR (%) | TPR @ 0.1% FPR (%) |
|---|---|---|---|
| Loss Thresholding (LOSS) | $0.640 \pm 0.003$ | $2.31 \pm 0.26$ | $0.34 \pm 0.08$ |
| Offline LiRA | $0.801 \pm 0.003$ | $4.36 \pm 0.51$ | $0.43 \pm 0.26$ |
| Online LiRA | $0.751 \pm 0.004$ | $2.00 \pm 0.22$ | $0.22 \pm 0.05$ |

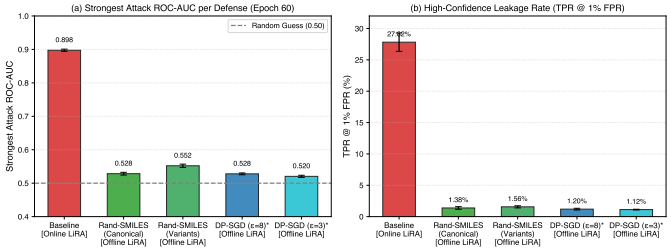



Fig. 3. Defense Comparison at Epoch 60 showing the strongest attack per defense. *Note on DP rows: preliminary run with possible loss of generative utility (not evaluated); an attack AUC near 0.5 may reflect an under-trained model and is not evidence of a useful defense.

### 4.5 Defense Comparison at Epoch 60

Table 4 and Fig. 3 compare two defensive approaches at epoch 60: (1) Randomized-SMILES training (E5a), where models are trained with random-SMILES data augmentation applied per batch; and (2) DP-SGD targeting $\epsilon \in \{8.0, 3.0\}$ with $\delta = 10^{-5}$.

Randomized-SMILES training reduces canonical attack performance: offline LiRA drops from 0.880 to $0.528 \pm 0.004$, and TPR@1% drops from 27.15% to 1.38%. When attacked via randomized variants scoring, offline LiRA AUC remains low at $0.552 \pm 0.005$.

In preliminary runs, DP-SGD reduces attack AUC near random guess ($0.500 - 0.528$). However, these models exhibited possible loss of generative utility (not evaluated): we did not evaluate the generative utility of the DP models; an attack AUC near 0.5 may reflect an under-trained model and is not evidence of a useful defense. No privacy-utility conclusion can be drawn without preserving generation validity. An attack AUC near 0.5 can come from a defense, an under-trained model, or a model that no longer generates valid molecules, so privacy metrics should be reported together with generative utility; this is a methodological caution, not a measured finding of this paper.

### 4.6 Generative Memorization Audit

To audit whether models generate verbatim copies of training leads, we sampled 5,000 molecules per model for three baseline models (Models 0, 15, 31; total 15,000 samples) on Apple Silicon (MPS). Generated molecules were evaluated for chemical

TABLE 4
Comparison of privacy defenses at epoch 60 under standard and calibrated attacks. All values use nll_sum (except randomized variants scoring which evaluates mean NLL across up to 10 distinct SMILES strings per molecule). The strongest attack per defense is bolded. *Note on DP rows: preliminary run with possible loss of generative utility (not evaluated); an attack AUC near 0.5 may reflect an under-trained model and is not evidence of a useful defense.

Panel (a): Calibrated Likelihood-Ratio Attacks (LiRA)

| | | Offline LiRA | | Online LiRA | | Strongest |
|---|---|---|---|---|---|---|
| Defense Configuration | Shadows | AUC | TPR@1% (%) | AUC | TPR@1% (%) | Attack |
| Baseline (Unmitigated) | 32 | $0.880 \pm 0.004$ | $27.15 \pm 1.63$ | $0.898 \pm 0.003$ | $27.82 \pm 1.47$ | Online LiRA |
| Rand-SMILES (Canonical) | 32 | $0.528 \pm 0.004$ | $1.38 \pm 0.21$ | $0.508 \pm 0.004$ | $1.09 \pm 0.17$ | Offline LiRA |
| Rand-SMILES (Variants, $\leq 10$ strings) | 32 | $0.552 \pm 0.005$ | $1.56 \pm 0.17$ | $0.519 \pm 0.005$ | $1.14 \pm 0.18$ | Offline LiRA |
| DP-SGD ($\epsilon = 8.0, \delta = 10^{-5}$)* | 8; prelim. | $0.528 \pm 0.003$ | $1.20 \pm 0.14$ | $0.501 \pm 0.005$ | $1.09 \pm 0.16$ | Offline LiRA |
| DP-SGD ($\epsilon = 3.0, \delta = 10^{-5}$)* | 8; prelim. | $0.520 \pm 0.003$ | $1.12 \pm 0.06$ | $0.500 \pm 0.002$ | $1.01 \pm 0.10$ | Offline LiRA |

Panel (b): Uncalibrated Loss Thresholding Baseline

| Defense Configuration | Shadows | LOSS AUC | LOSS TPR@1% (%) |
|---|---|---|---|
| Baseline (Unmitigated) | 32 | $0.689 \pm 0.005$ | $2.96 \pm 0.24$ |
| Rand-SMILES (Canonical) | 32 | $0.506 \pm 0.004$ | $1.08 \pm 0.16$ |
| Rand-SMILES (Variants, $\leq 10$ strings) | 32 | $0.506 \pm 0.004$ | $1.10 \pm 0.16$ |
| DP-SGD ($\epsilon = 8.0, \delta = 10^{-5}$)* | 8; prelim. | $0.505 \pm 0.004$ | $1.13 \pm 0.15$ |
| DP-SGD ($\epsilon = 3.0, \delta = 10^{-5}$)* | 8; prelim. | $0.503 \pm 0.004$ | $1.10 \pm 0.06$ |

validity, uniqueness, and exact string matches against 10,000 training members versus 10,000 holdout molecules (Table 5 and Fig. 4).

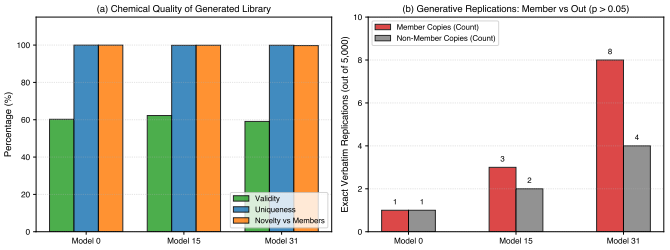



Fig. 4. Generative Memorization and Chemical Quality Audit (E6) across 15,000 samples.

Across 15,000 samples, models produced 12 exact member copies and 7 holdout copies. Individual binomial test $p$-values are 0.750, 0.500, and 0.194. The pooled one-sided binomial test yields $p = 0.1796$ ($p > 0.05$), and a Fisher exact test yields $p = 0.359$. Under the evaluated sample budget (5,000 per model), the difference between member copies and holdout copies is not statistically significant. Generative memorization cannot be claimed as statistically confirmed without higher-power sampling ($\geq$ 100k samples).

## 5 Related Work

Our work interfaces with privacy auditing, generative chemistry, and membership inference:

- Membership Inference Attacks: Carlini et al. [4] established Likelihood-Ratio Attacks (LiRA) to overcome sequence hardness confounding. Shokri et al. [2] originally formulated shadow-model attacks. Recent theoretical analysis by Zhang et al. [7] emphasized that empirical membership inference attacks face significant evidentiary caveats and cannot formally prove that a specific sample was present in a training set without known reference baselines.
- Privacy in Cheminformatics & Molecular Models: Krüger et al. [8] investigated membership leakage in molecular property predictors (molprivacy), demonstrating that publishing neural network weights poses data confidentiality risks. Aubinais et al. [9] studied membership inference risks in quantized models, incorporating chemical language model experiments. Ross et al. [10] developed GP-MoLFormer and noted training data memorization across 1.1 billion SMILES strings.
- Privacy Auditing in Biological Sequence Models: Nemecek et al. [11] analyzed memorization and privacy risk in genomic language models, illustrating domain-specific sequence leakage dynamics.
- SMILES Randomization & Data Augmentation: Arús-Pous et al. [12] and Bjerrum [13] demonstrated that randomized SMILES enumeration expands the effective representation space and regularizes generative models against overfitting.
- Differential Privacy in Deep Learning: Abadi et al. [14] and the Opacus library [6] formalized differential privacy via DP-SGD and Rényi Differential Privacy (RDP) accounting.

Our contribution provides a focused empirical audit of small causal SMILES language models across training epochs, dataset sizes, randomized representation scoring, and DP-SGD training. We do not claim to present the first or definitive study, but rather a transparent, reproducible empirical audit of these dynamics in small chemical generative LMs.

## 6 Limitations

This preliminary study has several explicit limitations:

1) Model Capacity and Architecture: Models evaluate a 2-layer, $d = 128$, $\sim$ 0.4M parameter GPT with mod-

TABLE 5
Generative memorization audit (Experiment E6; 5,000 samples per model evaluated against 10,000 training members and 10,000 non-members). Replication counts denote verbatim matches: Count = Fraction $\times$ 5,000. One-sided binomial tests ($p = 0.5$) evaluate whether member replications exceed non-member replications. All $p$-values exceed 0.05, and tests do not show a statistically significant replication difference at the current sample size.

| Model ID | Validity (%) | Uniqueness (%) | Novelty (%) | Member Copies | Out Copies | Binomial $p$-value | Statistically Significant? |
|---|---|---|---|---|---|---|---|
| Model 0 | 60.24% | 99.97% | 99.97% | 1 | 1 | 0.7500 | No ($p > 0.05$) |
| Model 15 | 62.26% | 99.87% | 99.90% | 3 | 2 | 0.5000 | No ($p > 0.05$) |
| Model 31 | 59.10% | 99.90% | 99.73% | 8 | 4 | 0.1938 | No ($p > 0.05$) |
| Pooled (15k) | 60.53% | 99.91% | 99.87% | 12 | 7 | 0.1796 | No ($p > 0.05$) |

est chemical validity ($\sim 60\%$). These findings may not generalize directly to larger generative models.

2) Dataset Scope: Experiments are restricted to a single filtered subset of the MOSES benchmark. Diverse proprietary libraries with differing scaffold distributions may exhibit different leakage profiles.
3) Shadow Ensemble Scale Asymmetry: While baseline and E5a arms used 32 shadow models, compute limits restricted E3, DP-SGD, and pretrain-FT to 8 shadow models, leading to higher estimation variance.
4) Reported Uncertainty: Metrics report standard deviation across leave-one-out shadow target models, rather than per-molecule bootstrap confidence intervals. Our computed 95% bootstrap CIs across molecules for epoch 60 online LiRA on three targets span roughly $\pm 0.005$ in AUC and $\pm 0.03$ in TPR@1% (raw per-molecule scoring parquets are omitted from the release repository due to storage constraints and are available on request).
5) Generative Sample Size and Hardware: Experiment E6 sampled 5,000 molecules per model (15,000 total) and ran on Apple Silicon (MPS), whereas shadow training was conducted on CUDA Tesla T4. Statistical tests do not show a significant difference at this sample size ($p > 0.05$).
6) DP-SGD Generative Utility: The DP-SGD arm represents a preliminary run with possible loss of generative utility (not evaluated): chemical validity evaluations were not recorded in the released summary, and we did not evaluate the generative utility of the DP models. An attack AUC near 0.5 may reflect an under-trained model and is not evidence of a useful defense. Furthermore, privacy accounting logs and accountant state deserialization were not independently verified.
7) Lack of From-Scratch Control: The pretrain $\rightarrow$ fine-tune experiment lacks an exact from-scratch baseline trained for 20 epochs on 10,000 molecules, preventing full disentanglement of pretraining effects from training duration.
8) Randomized-SMILES Utility Evaluation: Validation NLL and generative chemical validity of randomized-SMILES-trained models were not evaluated.

## 7 Conclusion

This preliminary study audited membership leakage in small SMILES language models. We observed that under canonical queries and likelihood access, models overfit over 60 epochs, enabling calibrated LiRA attacks to reach an AUC of 0.898. In our setting, scoring with mean NLL over up to 10 string forms (canonical included) removes most of the signal; training with randomized SMILES also reduces attack AUC to $0.51-0.56$. While DP-SGD is designed to bound the influence of individual training examples, preliminary implementations exhibited possible loss of generative utility (not evaluated); an attack AUC near 0.5 may reflect an under-trained model and is not evidence of a useful defense. Future work must investigate scaling laws on larger architectures, higher-power generative sampling, and privacy-preserving training that maintains chemical validity.

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